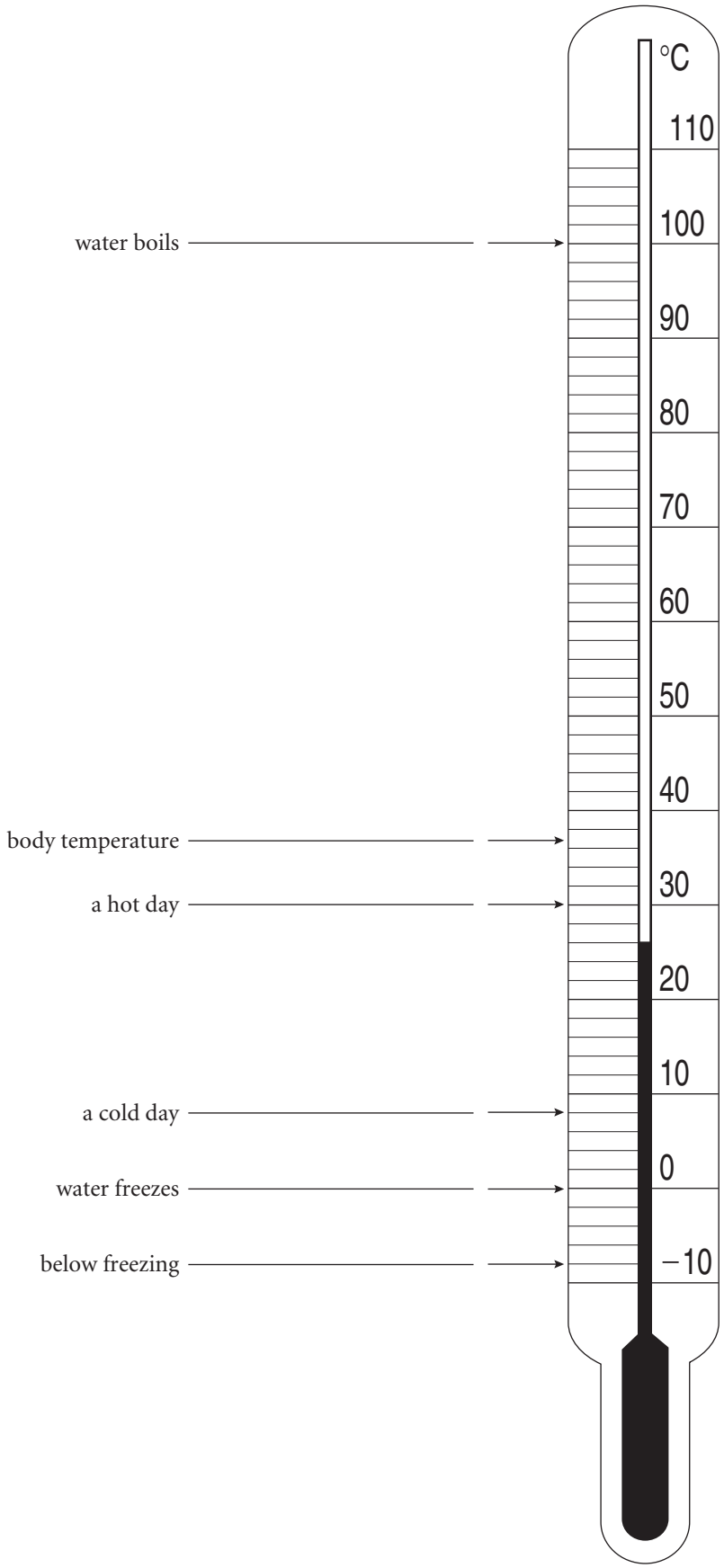
What's the temperature?

On the diagram of a thermometer, write in the temperature readings.



© V Newman, 2002, The Heinemann Science Scheme

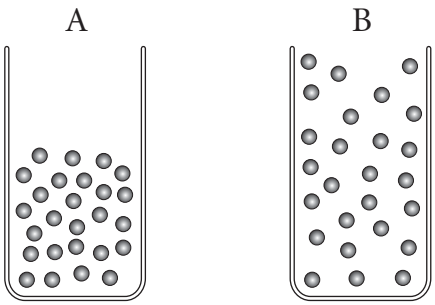
Warming up and cooling down

1 Complete the sentences below. Use these words to fill in the gaps.

insulation	Bunsen	lid	temperature	fell

When we heated some water in a beaker, we used a _____ burner. As the water was heated, its _____ rose. When we stopped heating, the temperature of the water _____.

2 The two diagrams show the particles in a gas at different temperatures. Fill in the gaps in the sentences below.



The gas in diagram _____ is at a lower temperature.

The particles are _____ together.

They have _____ energy.

The gas in diagram _____ is at a higher temperature.

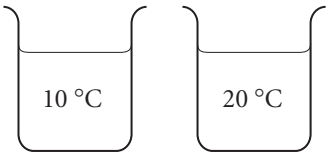
The particles are _____ _____.

They have _____ energy.

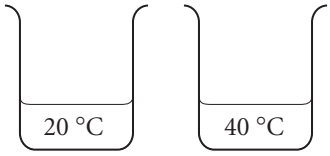
© V Newman, 2002, The Heinemann Science Scheme

Temperature and heat energy

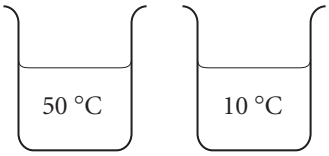
1 The beakers of water in each pair are poured together. Say what you think the temperature will be afterwards for each pair.



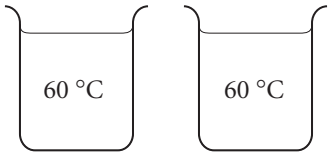
a _____ °C



b _____ °C

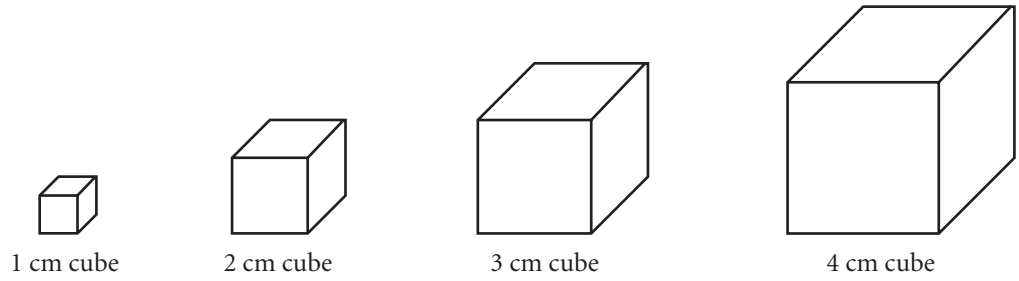


c _____ °C



d _____ °C

2 These four cubes of copper are at room temperature. They are put in a saucepan of boiling water.



a Which one takes the longest time to heat up to 100 °C?

b Which one is the quickest to heat up to 100 °C?

© V Newman, 2002, The Heinemann Science Scheme

Expanding and contracting

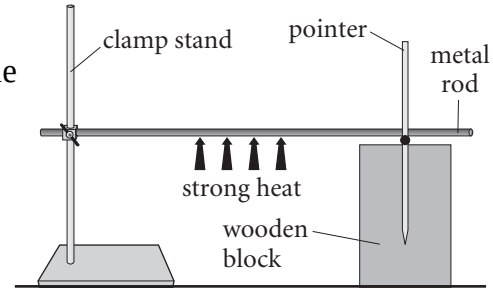
The diagrams show experiments to demonstrate expansion. Complete each description by filling in the gaps in the sentences below.

1 When the metal rod was _____,

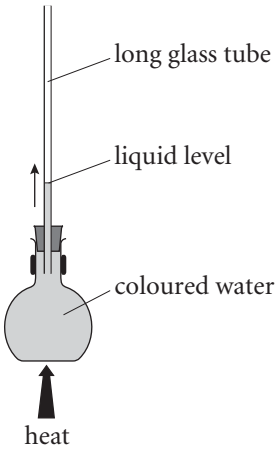
the pointer moved clockwise. This was because the metal rod _____ as it became hotter.

As the rod cooled it _____,

so the pointer moved back the other way.



2

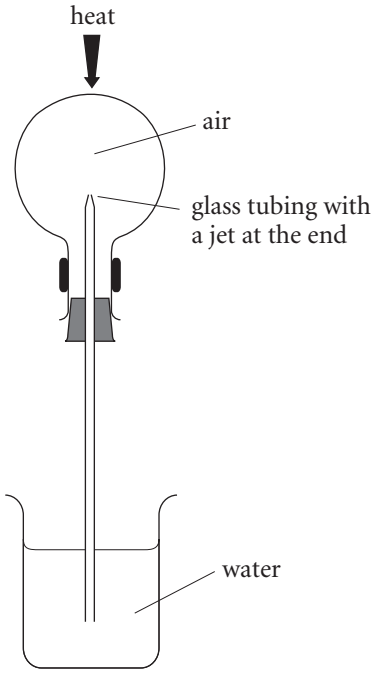


As the liquid was heated, it _____ and we could see it move up the tube. As the liquid cooled, it _____ so the liquid level in the tube went down.

3 When the flask was heated, _____

came out of the end of the tube in the water.

This shows that when the _____ in the flask was heated, it _____.



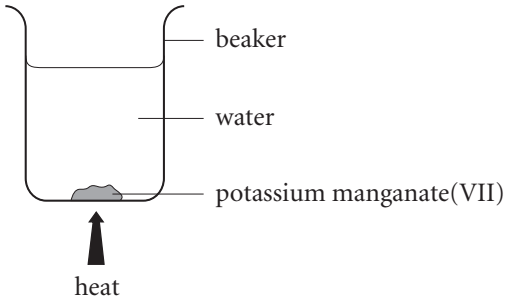
4 Which expands the most when you heat it, a solid, a liquid or a gas? _____

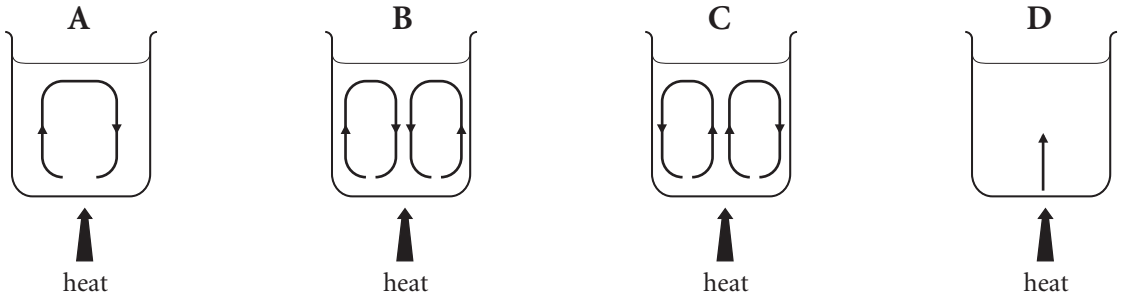
© V Newman, 2002, The Heinemann Science Scheme

Convection

1 The diagram shows an experiment you could use to demonstrate convection in liquids.

As the experiment continues, a convection current starts in the beaker. Which beaker below shows how the convection current would look?





The correct beaker is _____.

2 Complete the following sentences. Use the words below to fill in the gaps. You may use each word once or more than once.

apart	dense	dissolves	energy	expand	rises

The coloured crystal _____ in the water.

The coloured water moves, showing that warm water _____.

When the water is warmed, the particles gain more _____ so they move further _____.

This makes the warm water _____. This is why it _____.

3 Complete the following sentences. Use the words below to fill in the gaps.

air	convection	expands	dense	rises

An electric heater heats the whole room. This is because the air above the heater is warmed, so it _____ and _____ because it is less _____. Cold _____ takes the place of warm air and so _____ currents are set up.

© V Newman, 2002, The Heinemann Science Scheme

Radiation and energy escape

1 Complete the following sentences. Use the words below to fill in the gaps.

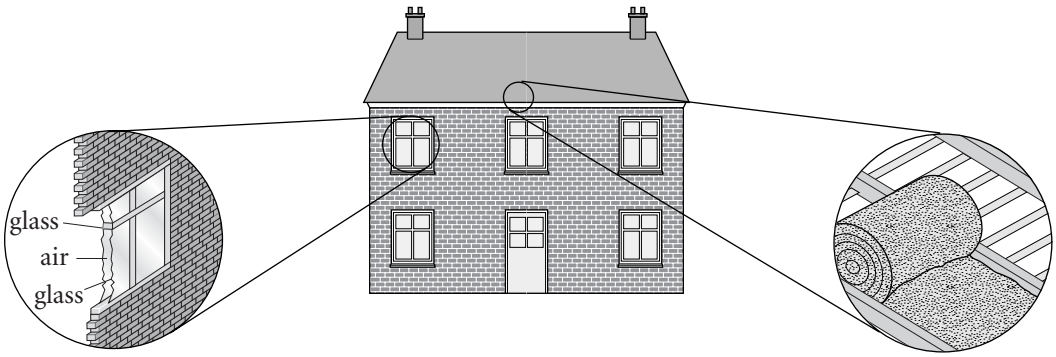
infrared radiation	**light**	**medium**	**vacuum**

Heat travels through space from the Sun to the Earth. This is an example of heat travelling by _____ _____.

This type of heat transfer requires no _____ – it can travel through a _____.

Another type of energy that travels like this is _____.

2



double glazing loft insulation

Complete the following sentences. Use the words below to fill in the gaps. You may use each word once or more than once.

air	**conductor**	**convection**	**insulator**

Double glazing is designed to reduce heat loss from a house. The air between the sheets of glass is a poor _____, and the layer of air is too narrow for _____ currents to be set up.

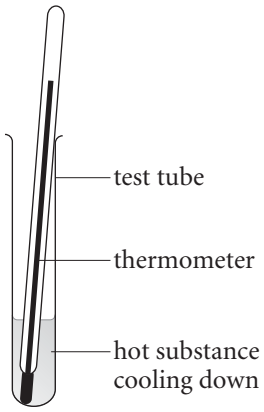
Loft insulation works by trapping pockets of _____ between the fibres. The trapped air is a good _____ and because it is trapped it reduces heat loss by _____.

© V Newman, 2002, The Heinemann Science Scheme

Explaining changes of state

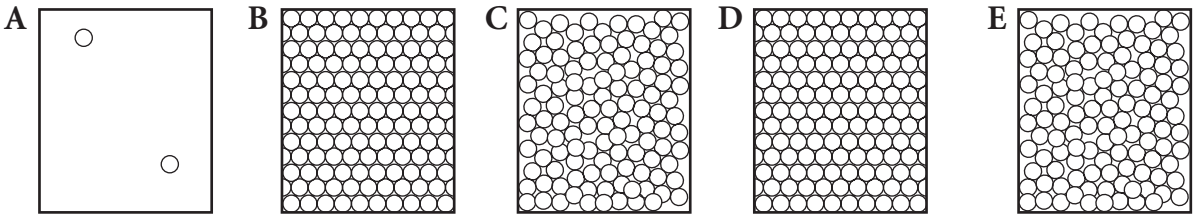
1 Some white solid was heated in a water bath. When it got hot, it melted and became a liquid.

The diagram shows the equipment for taking the temperature of the liquid. Fill in the gaps in the sentences below to describe what happened next.



The liquid was then allowed to _____ down. At first, its temperature went down. Then at about 50 °C the temperature stayed _____ for some minutes. At this temperature the liquid was changing into a _____.

2 The diagrams below show the particles in a solid, a liquid and a gas. Write the letters in the correct order for the diagrams to show a solid being heated, eventually becoming a gas and then cooling down and becoming a solid again.



The correct order is _____ _____ _____ _____ _____

3 Fill in the gaps in these sentences.

a At 0 °C, water forms _____. This temperature is called the _____ _____ of water.

b At 100 °C, water forms _____. This temperature is called the _____ _____ of water.

© V Newman, 2002, The Heinemann Science Scheme